Light

GCSE Computer Science - Programming Data Structures (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which statement best describes why arrays are useful in programming?

- ○ **A.** Arrays allow a single variable name to store and manage multiple related values of the same type
- ○ **B.** Arrays make programs run twice as fast as using individual variables
- ○ **C.** Arrays can only store exactly three values at a time
- ○ **D.** Arrays automatically sort data into alphabetical order

Q2 (1 mark)

What is each individual value stored in an array called?

- ○ **A.** A field
- ○ **B.** An element
- ○ **C.** A cell
- ○ **D.** A record

Q3 (1 mark)

An array called `scores` has 10 elements. Using zero-based indexing, what would happen if you tried to access `scores[10]`?

- ○ **A.** It would return the last element in the array
- ○ **B.** It would return the value 10
- ○ **C.** It would cause an index out of bounds error because valid indices are 0 to 9
- ○ **D.** It would return the first element in the array

Q4 (1 mark)

What is the key difference between arrays and records?

- ○ **A.** Records use named fields that can hold different data types, while arrays use numeric indices for items of the same type
- ○ **B.** Arrays can store data but records cannot
- ○ **C.** Arrays can only store numbers but records can store text
- ○ **D.** There is no difference - they are two names for the same thing

Q5 (1 mark)

Which real-world structure is most similar to a two-dimensional array?

- ○ **A.** A single queue of people waiting in line
- ○ **B.** A stack of plates
- ○ **C.** A dictionary with word definitions
- ○ **D.** A spreadsheet with rows and columns

Q6 (1 mark)

After defining a record type called `Employee`, what do you create when you make a variable of that type?

- ○ **A.** An instance of the record
- ○ **B.** A copy of the record definition
- ○ **C.** A new programming language
- ○ **D.** A two-dimensional array

Q7 (1 mark)

If an array has a length of 8, how many elements does it contain?

- ○ **A.** 8 elements (indices 0 to 7)
- ○ **B.** 7 elements (indices 1 to 7)
- ○ **C.** 9 elements (indices 0 to 8)

- ○ **D.** 8 elements (indices 1 to 8)

Q8 (1 mark)

In the record definition below, what data type is the `price` field?

```
RECORD Product
    name: String
    price: Real
    quantity: Integer
ENDRECORD
```

- ○ **A.** String
- ○ **B.** Real
- ○ **C.** Integer
- ○ **D.** Product

Q9 (1 mark)

Which statement correctly changes the third element of an array called `data` to the value 50?

- ○ **A.** data[2] <- 50
- ○ **B.** data[3] <- 50
- ○ **C.** data.3 <- 50
- ○ **D.** third(data) <- 50

Q10 (1 mark)

In a 2D array called `grid`, what does the expression `grid[0][0]` access?

- ○ **A.** The element in the first row and first column
- ○ **B.** The element in the second row and second column
- ○ **C.** All elements in the first row
- ○ **D.** The total number of elements in the grid

Q11 (1 mark)

Why would a programmer choose to use a record instead of separate variables?

- ○ **A.** Records use less memory than variables
- ○ **B.** Records automatically validate all data entered into them
- ○ **C.** Records can only store one piece of data
- ○ **D.** Records keep related data together as a single logical unit, making code more organised

Q12 (1 mark)

What values does this array contain after the declaration?

```
numbers <- [7, 14, 21, 28]
```

- ○ **A.** The values 1, 2, 3, 4

- ○ **B.** The values 0, 1, 2, 3
- ○ **C.** The values 7, 14, 21, 28
- ○ **D.** No values until elements are added individually

Q13 (2 marks)

What is output by this code?

```
items <- ["apple", "banana", "cherry"]
FOR i <- 0 TO 2
    OUTPUT items[i]
ENDFOR
```

- ○ **A.** apple, banana, cherry
- ○ **B.** 0, 1, 2
- ○ **C.** cherry, banana, apple
- ○ **D.** apple only

Q14 (2 marks)

What is the value of `result` after this code executes?

```
nums <- [4, 8, 2, 6]
result <- nums[0] + nums[2]
```

- ○ **A.** 12
- ○ **B.** 10

- C. 6
- D. 14

Q15 (2 marks)

What is output by this code?

```
RECORD Pet
    species: String
    name: String
    age: Integer
ENDRECORD

myPet <- Pet("Cat", "Whiskers", 3)
OUTPUT myPet.name
OUTPUT myPet.age
```

- A. Cat and 3
- B. Whiskers and 3
- C. Pet and Whiskers
- D. name and age

Q16 (2 marks)

A game stores a 3x3 board as a 2D array:

```
board <- [["X", "O", "X"],
          ["O", "X", "O"],
          ["O", "X", "X"]]
```

What value is stored at `board[2][0]`?

- A. X
- B. 2
- C. The entire third row
- D. O

Q17 (2 marks)

What is output by this code?

```
grades <- ["A", "B", "A", "C", "A", "B"]
count <- 0
FOR i <- 0 TO 5
    IF grades[i] = "A" THEN
        count <- count + 1
    ENDIF
ENDFOR
OUTPUT count
```

- A. 6
- B. 3
- C. 2
- D. 1

Q18 (2 marks)

After this code executes, what is the value of `account.balance`?

```
RECORD BankAccount
    holder: String
    balance: Real
```

```
ENDRECORD

account <- BankAccount("John Smith", 500.00)
account.balance <- account.balance + 150.00
```

- A. 500.00
- B. 150.00
- C. 650.00
- D. 350.00

Q19 (2 marks)

What is output by this code that finds the smallest value?

```
values <- [15, 8, 23, 4, 19]
smallest <- values[0]
FOR i <- 1 TO 4
    IF values[i] < smallest THEN
        smallest <- values[i]
    ENDIF
ENDFOR
OUTPUT smallest
```

- A. 15
- B. 0
- C. 8
- D. 4

Q20 (2 marks)

A 2D array is declared as:

```
table <- [[1, 2, 3, 4],
          [5, 6, 7, 8],
          [9, 10, 11, 12]]
```

How many rows and columns does this array have?

- ○ **A.** 4 rows and 3 columns
- ○ **B.** 3 rows and 4 columns
- ○ **C.** 12 rows and 1 column
- ○ **D.** 2 rows and 6 columns

Q21 (2 marks)

This code outputs array elements. In what order are they displayed?

```
letters <- ["P", "Q", "R", "S"]
FOR i <- 3 DOWNTO 0
    OUTPUT letters[i]
ENDFOR
```

- ○ **A.** P, Q, R, S
- ○ **B.** S, R, Q, P
- ○ **C.** 3, 2, 1, 0
- ○ **D.** S only

Q22 (2 marks)

A program needs to store information about 50 books, where each book has a title, author, and ISBN. What is the most appropriate data structure?

- ○ **A.** Three separate arrays: one for titles, one for authors, one for ISBNs
- ○ **B.** A single record with 150 fields
- ○ **C.** An array of 50 Book records, where each record contains title, author, and ISBN fields
- ○ **D.** A single string variable containing all the information

Q23 (2 marks)

An array `names <- ["Ali", "Ben", "Cat", "Dan"]` stores student names. What is the result of `LENGTH(names[2])`?

- ○ **A.** 4
- ○ **B.** 2
- ○ **C.** 3
- ○ **D.** Cat

Q24 (3 marks)

What is output by this code?

```
matrix <- [[2, 4],
           [6, 8],
           [10, 12]]
sum <- 0
FOR row <- 0 TO 2
    sum <- sum + matrix[row][1]
```

```
ENDFOR
OUTPUT sum
```

- A. 18
- B. 6
- C. 42
- D. 24

Q25 (3 marks)

What is output by this code?

```
RECORD Pupil
    name: String
    score: Integer
ENDRECORD

class <- [Pupil("Emma", 85), Pupil("Finn", 72), Pupil("Grace", 91
highest <- class[0].score
FOR i <- 1 TO 2
    IF class[i].score > highest THEN
        highest <- class[i].score
    ENDIF
ENDFOR
OUTPUT highest
```

- A. 85
- B. 72
- C. 91
- D. Grace

Q26 (3 marks)

What are the contents of the array after this code executes?

```
data <- [10, 20, 30, 40, 50]
FOR i <- 0 TO 3
    data[i] <- data[i + 1]
ENDFOR
data[4] <- 0
```

- ○ **A.** [10, 20, 30, 40, 50]
- ○ **B.** [10, 20, 30, 40, 0]
- ○ **C.** [0, 10, 20, 30, 40]
- ○ **D.** [20, 30, 40, 50, 0]

Q27 (3 marks)

What is output by this code that processes a 2D array?

```
grid <- [[0, 1, 0],
         [1, 1, 1],
         [0, 1, 0]]
count <- 0
FOR r <- 0 TO 2
    FOR c <- 0 TO 2
        IF grid[r][c] = 1 THEN
            count <- count + 1
        ENDIF
    ENDFOR
ENDFOR
OUTPUT count
```

- ○ **A.** 9

- B. 4
- C. 5
- D. 3

Q28 (3 marks)

A cinema booking system needs to store information about each screening: film title, screen number (1-8), start time, and whether subtitles are shown. Analyse which record definition is most appropriate.

- A. RECORD Screening filmTitle: String screenNumber: Integer startTime: String hasSubtitles: Boolean ENDRECORD
- B. RECORD Screening filmTitle: Integer screenNumber: String startTime: Boolean hasSubtitles: Real ENDRECORD
- C. RECORD Screening allData: String ENDRECORD
- D. ARRAY Screening[filmTitle, screenNumber, startTime, hasSubtitles]

Q29 (3 marks)

What is output by this code?

```
temps <- [18, 22, 25, 19, 16]
total <- 0
FOR i <- 0 TO 4
    total <- total + temps[i]
ENDFOR
average <- total / 5
OUTPUT average
```

- A. 100

- B. 5
- C. 25
- D. 20

Q30 (3 marks)

A spreadsheet stores weekly sales data where rows represent 4 salespeople and columns represent 5 weekdays. Analyse what `sales[3][4]` represents.

- A. The total sales for salesperson 3
- B. The sales figure for the fourth salesperson on Friday
- C. The sales figure for the third salesperson on Thursday
- D. All sales figures for Friday

Total: **55** marks • Time: **30 min**

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